Project: Worked examples
Title: Helical springs
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1. Scope and method

Calculates the rate, working points, stresses, buckling, fatigue and surge of a cold-coiled round-wire helical spring: compression, extension or torsion. The characteristic is linear, with hook stresses for extension and legs for torsion springs.

WHAT IT CALCULATES	RELATION
Index and rate	$C = D/d$; $k = G \cdot d^4/(8 \cdot D^3 \cdot n_{active})$ (compression, extension); $k_\varphi = E \cdot d^4/(64 \cdot D \cdot n_a)$, $n_a = n_{active} + (l_{leg,1} + l_{leg,2})/(3\pi \cdot D)$ (torsion)
Spring stress	$\tau_k = k_c \cdot 8 \cdot F \cdot D/(\pi \cdot d^3)$, $k_c = (C + 0.5)/(C - 0.75)$ (Bergsträsser) or $(4C - 1)/(4C - 4) + 0.615/C$ (Wahl); torsion $\sigma = q \cdot 32 \cdot M/(\pi \cdot d^3)$ with $q = (C + 0.07)/(C - 0.75)$ (EN 13906-3, the default) or $K_i = (4C^2 - C - 1)/(4C \cdot (C - 1))$ (Shigley)
Solid length (compression)	$n_t = n_{active} + e_n$; EN 13906-1 form $L_s = d_{max} \cdot (n_t + e_s)$ with $(e_n, e_s) = (0, 1)$ plain, $(1, 0)$ plain and ground, $(2, 1.5)$ squared (closed, unground), $(2, 0)$ squared and ground, and the minimum coil clearance $S_a = (0.0015 \cdot D^2/d + 0.1 \cdot d) \cdot n_{active}$ checked at the working length ($L_2 \geq L_s + S_a$); Shigley form $L_s = d \cdot (n_t + e_s)$ with $e_s = 1$ for squared; pitch $p = (L_o - a \cdot d)/(n_{active} + b)$
Extension spring hook stresses	$F = F_o + k \cdot s$; hook bending $\sigma_A = F \cdot [K_A \cdot 16 \cdot D/(\pi \cdot d^3) + 4/(\pi \cdot d^2)]$, $K_A = (4C_1^2 - C_1 - 1)/(4C_1 \cdot (C_1 - 1))$, $C_1 = 2r_1/d$; hook torsion $\tau_B = K_B \cdot 8 \cdot F \cdot D/(\pi \cdot d^3)$, $K_B = (4C_2 - 1)/(4C_2 - 4)$, $C_2 = 2r_2/d$
Buckling (compression)	$\lambda_{eff} = \nu \cdot L_o/D$; $y_{cr} = L_o \cdot C_1' \cdot [1 - \sqrt{(1 - C_2'/\lambda_{eff}^2)}]$, $C_1' = E/(2(E - G))$, $C_2' = 2\pi^2(E - G)/(2G + E)$; stable at any deflection when $\lambda_{eff}^2 < C_2'$
Fatigue (Goodman, as M4)	$1/n_f = a/S_e + m/S_u$ with $a = (x_2 - x_1)/2$, $m = (x_2 + x_1)/2$ of the stresses at the two working points; $S_u = f_{su} \cdot R_m$ in shear, R_m in bending; the extension hook has its own rows, bending σ_A with $S_{e,b}$ and R_m, side bend τ_B with S_e and $f_{su} \cdot R_m$; R_m typed or $R_m = A/d\text{^}m$ from a material record (A in MPa, d in mm)
Surge frequency	$f_n = (d/(2\pi \cdot n_{active} \cdot D^2)) \cdot \sqrt{(G/(2\rho))}$; warning when $f_n/f_{op} < 13$
Torsion body	$D_w = D \cdot n_{active}/(n_{active} + \varphi/2\pi)$; $L_w = d \cdot (n_{active} + 1 + \varphi/2\pi)$

Basis: Calculation of compression springs: rate, stress correction, solid length, buckling, fatigue, surge[1]; Calculation of extension springs: rate, initial tension, hook stress[2]; Calculation of torsion springs: rate, bending stress with curvature correction[3]; Helical springs: end geometry, buckling, critical frequency, extension and torsion springs[4](Ch. 10, 10th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Kind of spring	Compression		spec: Plunger assembly PLG-210 §4 rev B	any
corr	Stress correction factor k_c for the shear stress	Bergsträsser		spec: Company spring standard SPR-01 (EN 13906-1 correction factor)	any
ends	End type (compression)	Squared and ground		drawing: PLG-210-04 rev B	any
work	Working points given as	Loads		spec: Plunger assembly PLG-210 §4 rev B	any
rm_{source}	Where $R_m(d)$ comes from: typed (no default: the wire certificate or your wire standard) or the curve R_m = A/d^m of a material record	Typed R_m(d)		assumption: R_m(d) typed for this example; a W3 material with the curve R_m = A/d^m can be applied instead.	any
$solid_{method}$	Solid length and clearance (compression): EN 13906-1 uses the maximum wire diameter, 1.5 d more for squared unground ends, and checks the minimum coil clearance at L_2; Shigley uses the nominal d	EN 13906-1 (d_max, coil clearance)		spec: Company spring standard SPR-01 (EN 13906-1 solid length and minimum coil clearance); no wire tolerance entered, so d_max is the nominal 3 mm	any
d	Wire diameter	3	mm	drawing: PLG-210-04 rev B	0.1 … 20 mm
D	Mean coil diameter	24	mm	drawing: PLG-210-04 rev B	0.5 … 400 mm
n_{active}	Active coils (torsion: coils of the body)	9		drawing: PLG-210-04 rev B	2 … 200
L0	Free length (compression)	70	mm	drawing: PLG-210-04 rev B	1 … 1000 mm
G	Shear modulus	79.3	GPa	assumption: Generic spring steel shear modulus; take it from the W3 material of the wire certificate.	20 … 100 GPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
E	Young's modulus	206	GPa	assumption: Generic spring steel Young's modulus; take it from the W3 material of the wire certificate.	50 … 250 GPa
ρ	Density	7850	kg/m³	assumption: Generic steel density.	1000 … 20000 kg/m³
S_e	Endurance strength for the stress kind (shear; bending in torsion) at N_c cycles, alternating (no default: from the wire maker or your fatigue reference)	450	MPa	assumption: Example alternating shear endurance strength at 10^7 cycles; take it from the wire maker or your fatigue reference (record spring.endurance).	50 … 2000 MPa
Rm	Tensile strength of the wire at this diameter, typed (no default)	1650	MPa	assumption: Example value for a 3 mm wire. Enter R_m(d) from your own copy of the wire standard or the wire certificate (record spring.wire-strength); the standard's table is not shipped.	300 … 3500 MPa
F_1	Force at working point 1 (loads)	60	N	spec: Plunger assembly PLG-210 §4 rev B	0 … 20000 N
F_2	Force at working point 2 (loads)	180	N	spec: Plunger assembly PLG-210 §4 rev B	0 … 20000 N
guided	Spring guided by a mandrel or in a bore (compression): the guide restrains it against buckling, so the free-buckling check is not made	no		default	any
ν_{seat}	End-support coefficient ν (0.5 both ends fixed, 0.7 one fixed one pinned, 1 both pinned, 2 one free)	0.5		spec: Plunger assembly PLG-210 §4: both end plates flat and parallel, no rotation	0.5 … 2
f_{op}	Operating (excitation) frequency	8	Hz	spec: Plunger assembly PLG-210 §4: maximum cycle rate	0 … 1000 Hz

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_{tau}	Permissible shear stress as a fraction of R_m (generic textbook default; replace with the licensed EN 13906 value)	0.45		assumption: Static permissible shear stress fraction of R_m for the example; take it from your design reference (record spring.permissible).	0.2 ... 0.8
f_{su}	Ultimate shear strength as a fraction of R_m (Goodman, shear; generic textbook default, replace with the licensed EN 13906 value)	0.67		assumption: Ultimate shear strength fraction of R_m used by the Goodman relation (textbook steel value); confirm in your reference.	0.4 ... 0.9
$n_{f,req}$	Required Goodman safety factor	1.2		spec: Company spring standard SPR-01	1 ... 5
N_c	Number of cycles the endurance strength belongs to	10000000		spec: Plunger assembly PLG-210 §4: duty life	10000 ... 10000000000

3. Assumptions

A1. The rate is constant between the working points, with no progressive or conical spring and no variable pitch. (Cold-coiled round wire up to 20 mm.)
A2. The active coils are as entered, and end coils count only in the solid length and the pitch. (Shigley 10th ed. Ch. 10, geometry of the ends.)
A3. Hook flexibility is not in the rate, and leg flexibility is in the rate only for torsion springs. (Shigley 10th ed. Ch. 10, geometry of the ends.)
A4. The EN 13906-1 solid length uses the maximum wire diameter d_{max} and adds 1.5 d for an unground squared end. (EN 13906-1 relations written in own notation, to be checked against the owner's copy, and Shigley 10th ed. Table 10-1.)
A5. The EN 13906-1 form checks the minimum coil clearance S_a at the working length L_2. (EN 13906-1 relation written in own notation, to be checked against the owner's copy.)
A6. Plain ends have no EN row and keep the geometric form, and the Shigley form (nominal d) is selectable. (Shigley 10th ed. Table 10-1.)
A7. The default bending stress factor of a torsion spring is the EN 13906-3 $q = (C + 0.07)/(C - 0.75)$. (EN 13906-3 relation written in own notation, to be checked against the owner's copy.)
A8. Shigley's $K_i = (4C^2 - C - 1)/(4C(C - 1))$ is selectable and 1 % lower at C = 8. (Shigley 10th ed. Ch. 10, torsion springs.)
A9. $R_m(d)$ and the endurance strength are user inputs from the wire standard or certificate, with no default. (No wire-strength table or permissible-stress diagram is shipped.)
A10. The permissible stress fractions default to generic textbook fractions, which the user replaces with the licensed EN 13906 value. (The defaults are 0.45 for f_{tau}, 0.75 for f_{sigma} and 0.67 for f_{su}.)
A11. $R_m(d)$ can come from the curve $R_m = A/d^m$ of a material record, A in MPa and d in mm. (The material is revision-pinned, and the user respects the diameter range of the fit.)
A12. The stress correction factor k_c is applied to the working and the solid stress alike. (Conservative for the static check. Choose 'None (uncorrected)' for the uncorrected static stress, and the fatigue stresses still carry the curvature factor, the Bergsträsser one when 'None' is chosen.)
A13. The hook shape beyond its two radii is not assessed in an extension spring. (Shigley 10th ed. Ch. 10,

extension springs.)
A14. Hook fatigue is Goodman, on the bending stress and on the side-bend torsion between the two working points. (The bending uses the endurance $S_{e,b}$, which has no default, and the side bend uses the shear endurance S_e.)
A15. The initial tension of an extension spring is a force the user enters. (Shigley 10th ed. Ch. 10, extension springs.)
A16. Surge is the first natural frequency of a spring with both ends fixed. (Shigley 10th ed. Ch. 10, critical frequency.)
A17. A spring with one end free has half the surge frequency. (Shigley 10th ed. Ch. 10, critical frequency.)
A18. rm_{source} (Where $R_m(d)$ comes from: typed (no default: the wire certificate or your wire standard) or the curve R_m = A/d^m of a material record) is taken as Typed $R_m(d)$. ($R_m(d)$ typed for this example; a W3 material with the curve R_m = A/d^m can be applied instead.)
A19. G (Shear modulus) is taken as 79.3 GPa. (Generic spring steel shear modulus; take it from the W3 material of the wire certificate.)
A20. E (Young's modulus) is taken as 206 GPa. (Generic spring steel Young's modulus; take it from the W3 material of the wire certificate.)
A21. ρ (Density) is taken as 7850 kg/m³. (Generic steel density.)
A22. S_e (Endurance strength for the stress kind (shear; bending in torsion) at N_c cycles, alternating (no default: from the wire maker or your fatigue reference)) is taken as 450 MPa. (Example alternating shear endurance strength at 10^7 cycles; take it from the wire maker or your fatigue reference (record spring.endurance).)
A23. Rm (Tensile strength of the wire at this diameter, typed (no default)) is taken as 1650 MPa. (Example value for a 3 mm wire. Enter $R_m(d)$ from your own copy of the wire standard or the wire certificate (record spring.wire-strength); the standard's table is not shipped.)
A24. f_{tau} (Permissible shear stress as a fraction of R_m (generic textbook default; replace with the licensed EN 13906 value)) is taken as 0.45. (Static permissible shear stress fraction of R_m for the example; take it from your design reference (record spring.permissible).)
A25. f_{su} (Ultimate shear strength as a fraction of R_m (Goodman, shear; generic textbook default, replace with the licensed EN 13906 value)) is taken as 0.67. (Ultimate shear strength fraction of R_m used by the Goodman relation (textbook steel value); confirm in your reference.)

4. Working

[7] C = D / d [1](13906-1:2013 / BS EN 13906-2:2013 / BS EN 13906-3:2014)
= 24 mm / 3 mm
= 8
Spring index

[15] k_c = (C + 0.5) / (C - 0.75)
= (8 + 0.5) / (8 - 0.75)
= 1.1724
Shear stress correction factor (Bergsträsser)

[16] $k_{c,fat}$ = k_c
= 1.1724
= 1.1724
Shear stress correction factor for the fatigue stresses (as the static stress)

[17] τ_{perm} = f_{tau} * Rm
= 0.45 × 1650 MPa
= 742.5 MPa
Permissible shear stress (fraction of R_m, user input)

[20] k = G * d^4 / (8 * D^3 * n_{active}) [1](13906-1:2013)
= 79.3 GPa × (3 mm)^4 / (8 × (24 mm)^3 × 9)
= 6.4535 N/mm
Spring rate

[21] $n_t = n_{active} + 2$
= 9 + 2
= 11
Total coils (derived: end coils added to the active coils)
[22] $d_{max} = d$
= 3 mm
= 3 mm
Maximum wire diameter (none entered: the nominal d)
[23] $L_s = d_{max} * (n_t + 0.0)$ [1](13906-1:2013)
= 3 mm × (11 + 0.0)
= 33 mm
Solid length L_c (EN 13906-1 form: maximum wire diameter)
[24] $S_a = (0.0015 * D^2 / d + 0.1 * d) * n_{active}$ [1](13906-1:2013)
= (0.0015 × (24 mm)^2 / 3 mm + 0.1 × 3 mm) × 9
= 5.292 mm
Minimum sum of the coil clearances S_a
[25] $L_n = L_s + S_a$ [1](13906-1:2013)
= 33 mm + 5.292 mm
= 38.292 mm
Minimum spring length $L_n = L_c + S_a$
[26] $p_{coil} = (L0 - 2 * d) / (n_{active} + 0)$
= (70 mm − 2 × 3 mm) / (9 + 0)
= 7.1111 mm
Pitch of the free spring (derived)
[27] $s_c = L0 - L_s$
= 70 mm − 33 mm
= 37 mm
Travel to solid
[28] $F_c = k * s_c$
= 6.4535 N/mm × 37 mm
= 238.78 N
Force at solid length
[32] $s_1 = F_1 / k$
= 60 N / 6.4535 N/mm
= 9.2974 mm
Deflection at working point 1
[33] $s_2 = F_2 / k$
= 180 N / 6.4535 N/mm
= 27.892 mm
Deflection at working point 2
[34] $\tau_1 = k_c * 8 * F_1 * D / (pi * d^3)$ [1](13906-1:2013)
= 1.1724 × 8 × 60 N × 24 mm / (π × (3 mm)^3)
= 159.23 MPa
Shear stress at working point 1
[35] $\tau_2 = k_c * 8 * F_2 * D / (pi * d^3)$ [1](13906-1:2013)
= 1.1724 × 8 × 180 N × 24 mm / (π × (3 mm)^3)
= 477.68 MPa
Shear stress at working point 2
[36] $\tau_c = k_c * 8 * F_c * D / (pi * d^3)$ [1](13906-1:2013)
= 1.1724 × 8 × 238.78 N × 24 mm / (π × (3 mm)^3)
= 633.67 MPa
Shear stress at solid length

[37] $u_2 = \tau_2 / \tau_{perm}$
= 477.68 MPa / 742.5 MPa
= 0.64335
Utilisation at working point 2
[38] $u_c = \tau_c / \tau_{perm}$
= 633.67 MPa / 742.5 MPa
= 0.85343
Utilisation at solid length
[39] $L_1 = L0 - s_1$
= 70 mm − 9.2974 mm
= 60.703 mm
Spring length at working point 1
[40] $L_2 = L0 - s_2$
= 70 mm − 27.892 mm
= 42.108 mm
Spring length at working point 2
[41] $u_s = s_2 / s_c$
= 27.892 mm / 37 mm
= 0.75384
Working deflection / travel to solid
[42] $s_n = L0 - L_n$
= 70 mm − 38.292 mm
= 31.708 mm
Travel to the minimum length L_n
[43] $u_n = s_2 / s_n$
= 27.892 mm / 31.708 mm
= 0.87965
Working deflection / travel to the minimum length (coil clearance at L_2)
[45] $\lambda_{eff} = v \cdot L_e / D$ [1](13906-1:2013)
= 0.5 × 70 / 24
= 1.4583
Effective slenderness
[46] C1p = E/(2(E − G)) [4](10th ed. Ch. 10, buckling)
= 206 / (2 × (206 − 79.3))
= 0.81294
Buckling constant C_1'
[47] $C2p = 2\pi^2(E - G)/(2G + E)$ [4](10th ed. Ch. 10, buckling)
= $2\pi^2$ × (206 − 79.3) / (2 × 79.3 + 206)
= 6.8595
Buckling constant C_2'
[48] u_b = stable at any deflection ($\lambda_{eff}^2 < C_2'$)
= 2.1267 < 6.8595
= 0
Deflection at point 2 against the buckling deflection (none: stable)
[49] f_n = d / (2 * pi * n_{active} * D^2) * sqrt(G / (2 * ρ)) [1](13906-1:2013 / BS EN 13906-2:2013)
= 3 mm / (2 × π × 9 × (24 mm)^2) × sqrt(79.3 GPa / (2 × 7850 kg/m^3))
= 207 Hz
Surge (natural) frequency, both ends fixed
[51] $r_f = f_n / f_{op}$
= 207 Hz / 8 Hz
= 25.875
Surge frequency / operating frequency

[56] $\tau_a = k_{c,fat} / k_c * (\tau_2 - \tau_1) / 2$
$= 1.1724 / 1.1724 \times (477.68\ MPa - 159.23\ MPa) / 2$
$= 159.23\ MPa$
Alternating shear stress
[57] $\tau_m = k_{c,fat} / k_c * (\tau_2 + \tau_1) / 2$
$= 1.1724 / 1.1724 \times (477.68\ MPa + 159.23\ MPa) / 2$
$= 318.46\ MPa$
Mean shear stress
[58] $S_{su} = f_{su} * Rm$
$= 0.67 \times 1650\ MPa$
$= 1105.5\ MPa$
Ultimate shear strength (fraction of R_m, user input)
[59] $u_f = \tau_a/S_e + \tau_m/S_{su}$ [4](10th ed. Ch. 6))
$= 159.23 / 450 + 318.46 / 1105.5$
$= 0.64191$
Goodman utilisation ($1/n_f$)
[60] $n_f = 1/u_f$
$= 1 / 0.64191$
$= 1.5579$
Goodman safety factor
[61] $u_{fat} = n_{f,req} * u_f$
$= 1.2 \times 0.64191$
$= 0.77029$
Fatigue utilisation against the required factor

5. Results

SYMBOL	VALUE	DESCRIPTION
k	6.453 N/mm	Rate
F_1	60 N	Force at point 1
F_2	180 N	Force at point 2
L_s	33 mm	Solid length
f_n	207.00 Hz	Surge frequency
C	8	Spring index
Rm	1650 MPa	Wire tensile strength
k_c	1.1724	Stress correction factor
n_t	11	Total coils
p_{coil}	7.111 mm	Pitch
s_1	9.297 mm	Deflection at point 1
s_2	27.89 mm	Deflection at point 2
L_1	60.70 mm	Length at point 1
L_2	42.11 mm	Length at point 2
s_c	37 mm	Travel to solid
F_c	238.8 N	Force at solid
τ_1	159.2 MPa	Stress at point 1
τ_2	477.7 MPa	Stress at point 2

SYMBOL	VALUE	DESCRIPTION
τ_c	633.7 MPa	Stress at solid
d_{max}	3 mm	Maximum wire diameter
S_a	5.292 mm	Minimum sum of coil clearances
L_n	38.29 mm	Minimum spring length
s_n	31.71 mm	Travel to L_n
y_{cr}	"— (stable at any deflection)"	Buckling deflection
buckling	"PASS"	Buckling
r_f	25.874	Surge ratio f_n/f_{op}
n_f	1.5578	Goodman safety factor
u_2	0.64	Utilisation at point 2
u_c	0.85	Utilisation at solid
u_s	0.75	Working deflection / travel to solid
u_n	0.88	Working deflection / travel to L_n (governing)
u_b	0.00	Deflection at point 2 / buckling deflection
u_f	0.64	Goodman utilisation
u_{fat}	0.77	Fatigue utilisation against the required factor

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Working deflection against the travel to the minimum length, keeping the coil clearance at L_2	27.89 mm	31.71 mm	0.88	[1](13906-1:2013)
PASS	Shear stress at solid length against the permissible stress	633.7 MPa	742.5 MPa	0.85	[1](13906-1:2013)
PASS	Goodman fatigue against the required safety factor	1.2	1.5579	0.77	[5](M4)
PASS	Working deflection against the travel to solid	27.89 mm	37 mm	0.75	[1](13906-1:2013)
PASS	Working deflection against the buckling deflection	stable at any deflection	$\lambda_{eff}^2 < C_2'$	—	[4](10th ed. Ch. 10, buckling)

7. Applicability

WARNING records: tabulated values not taken from a reference record are G, E, ρ (Material properties); Rm (Wire tensile strength $R_m(d)$); f_{tau}, f_{su} (Permissible and ultimate stress fractions); v_{seat} (Seating (end-support coefficient)); S_e (Endurance strength). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

```
Template mech.spring@1.6.0 Content hash e5989f67c99ce450 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. BS EN 13906-1:2013.
2. BS EN 13906-2:2013.
3. BS EN 13906-3:2014.
4. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
5. Goodman, as.